

Piyush Patle

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EDUCATION

Veermata Jijabai Technological Institute

B.Tech in Electrical Engineering, Minor in Robotics (CGPA-7.61/10)

Mumbai, India

Aug 2023 - Present

G.H. Raisonni Public School

CBSE, Class XII (84.8%)

Nagpur, India

2023

EXPERIENCE

Systems Intern, Mecha | Vadodara

May 2026 – Present

- **Project:** Linux board bring-up for the NXP i.MX95 platform.
- Applied upstream Linux kernel patch series from community developers to enable HDMI, camera, VPU, audio, and other peripherals.
- Validated HDMI, camera, and multimedia functionality using `modetest`, `media-ctl`, `libcamera`, and `GStreamer`, and resolved issues related to firmware, device tree, and kernel configuration.

Undergrad Research Intern, CDAC | Pune

June 2025 – Aug 2025

- **Project:** Porting the Lustre filesystem to Linux 6.12 on RISC-V.
- Built and tested NFS, BeeGFS, and Lustre for distributed storage and HPC workloads.
- Worked with Linux kernel components including VFS, `ext4`/`ldisksf`, kernel modules, and crash debugging.
- Resolved Linux 6.12 compatibility issues in Lustre, validated functionality on a RISC-V cluster.
- **Docs:** [Certificate](#) | [LOR](#) | [Research Paper](#)

Open-source Contributions

Remote

- **Linux Foundation:** Participated in the Linux Kernel Mentorship Program (LKMP); contributed patches that were merged into the mainline Linux kernel. ([Upstream Patches](#))
- **Apache NuttX:** Developed BSP support for PocketBeagle 2 (AM62x), improved documentation, and submitted refactoring patches. Multiple pull requests were merged upstream. ([Closed PRs](#))

PROJECTS

EMG-Armband | *Embedded C, ESP-NOW*

July 2024 – Oct 2024

- Built an EMG acquisition system using ESP32 and ESP-NOW for wireless real-time data streaming.
- Implemented gesture recognition for human-computer interaction and biofeedback applications.
- **Docs:** [Project Report](#) | [Blog](#)

PageForge | *C, Linux*

April 2026

- Implemented a simplified Linux-inspired memory management system in C, including a buddy allocator, slab allocator, and page table simulation.
- Built the project to understand Linux kernel memory allocation concepts and validated it using Unity tests and QEMU.
- **GitHub:** [Repository](#)

TECHNICAL SKILLS

Languages	Embedded C, C++
Operating Systems	Linux, FreeRTOS, ESP-IDF
Kernel/Embedded	Linux Kernel, Device Drivers, libcamera
Tools	Git, QEMU, GitHub Actions, Valgrind, Make
Hardware	STM32, ESP32, Raspberry Pi, BeagleBoard, NXP i.MX95

POSITIONS OF RESPONSIBILITY

Core Member | Society of Robotics and Automation (SRA)

Aug 2024 – Present

- * Conducted workshops on Embedded C, ESP32 robotics, OpenCV, and ROS 2 for WALL-E, PIXELS, and MARIO.
- * Mentored junior students on embedded systems, control systems, and low-level programming.
- * Mentored embedded projects like a Stewart platform ball balancer during Eklavya, SRA's 2-month mentorship program.

Technical Head | Technovanza, Technical Committee

Aug 2023 – Mar 2025

- * Led the technical committee of Technovanza, overseeing project builds, workshops, and event infrastructure; organized robotics workshops for 300+ students.
- * Spearheaded initiatives contributing to House Cup victory; designed and executed a technical workshop from scratch for 400+ students.

Marketing Manager | Swachh VJTI Eco-Friendly Action Club

Aug 2024 – Mar 2025